CPT202 Software Engineering
Coursework 1
Software Requirements Analysis

Student ID: 2361387

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1 Project Identification

Project Title:

Option C: An online project selection system for teachers to approve

Identified Project Modules:

- Module 1: User Registration and Authentication
- Module 2: Profile Management
- Module 3: Project Topic Creation, Editing, Publishing and Closing
- Module 4: Category Management
- Module 5: Project Search, Browsing and Request History Viewing
- Module 6: Application Submission, Tracking and Request Status Viewing
- Module 7: Application Processing and Allocation
- Module 8: Rule Management (Allocation Control)
- Module 9: System Dashboard (Global Overview)

Assigned Module:

Module 6: Application Submission, Tracking and Request Status Viewing

2 Primary Stakeholder and Operational Contexts

Primary Stakeholder

- Students who are participating in the project selection process — from submitting requests to tracking outcomes.

Operational Contexts

- When a student decides on a project topic, there is no formal channel to officially express interest to the supervising teacher.
- Students have limited visibility into the status of their request.
- Students who need to revise information in a submitted application cannot edit the request before the teacher reviews it.

- Students who no longer wish to continue with a submitted request cannot withdraw it before the teacher acts.
- Students receive no notifications when a teacher updates a request status, causing delays in follow-up response.

3 Pre-Groomed PBIs

Note: Only the first five PBIs will be marked.

PBI #	User Story	Acceptance Criteria
1	As a student who has selected a project topic, I want to submit a formal request with the required notes to the supervising teacher, so that I can officially express my interest in the project and enter the approval process.	• **Given** the student has selected a published project topic and entered the required notes, **when** the student submits the request, **then** the system will create the request, record the submission time, set its status to "Pending", and display a confirmation message. • **Given** the student has not selected a published project topic or has not entered the required notes, **when** the student attempts to submit the request, **then** the system will reject the submission and prompt the student to complete the missing requirement.
2	As a student who has submitted a project request, I want to view the current status of my request, so that I can stay informed about the teacher's decision without needing to follow up manually.	• **Given** a student has submitted one or more requests, **when** the student opens the request list, **then** the system will display each request together with its current status. • **Given** a student has not submitted any request, **when** the student opens the request list, **then** the system will indicate that no request is available. • **Given** a request status is displayed, **when** it appears in the system, **then** it will be one of the following: "Pending", "Approved", "Rejected", or "Withdrawn".

3	As a student who has submitted a project request, I want to edit my pending request before it is reviewed, so that I can correct or update the submitted information without creating a new application.	• **Given** the request status is "Pending", **when** the student edits and submits valid changes, **then** the system will save the updated information and keep the status as "Pending". • **Given** the request status is "Approved", "Rejected", or "Withdrawn", **when** the student attempts to edit the request, **then** the system will not allow the edit. • **Given** a pending request is edited successfully, **when** the student saves the changes, **then** the updated request will remain associated with the same request rather than creating a new one.
4	As a student who has submitted a project request, I want to withdraw my pending request before the teacher makes a decision, so that I can stop an application I no longer wish to continue.	• **Given** the request status is "Pending", **when** the student confirms the withdrawal, **then** the system will change the request status to "Withdrawn". • **Given** the request status is "Approved", "Rejected", or "Withdrawn", **when** the student attempts to withdraw the request, **then** the system will not allow the withdrawal. • **Given** a request has been withdrawn successfully, **when** the student views the request status, **then** the system will display the request as "Withdrawn".
5	As a student waiting for a teacher's decision, I want to receive a notification when the status of my request is updated, so that I can respond promptly without repeatedly checking the system manually.	• **Given** the status of a request changes, **when** the update is completed, **then** the system will send an in-application notification to the student. • **Given** a notification is displayed for a status update, **when** the student views it, **then** it will show the project title and the updated status.

4 Collaboration (Peer Review and Feedback)

Peer Reviewed:

Ziyi Jia / 2362573

Feedback Provided to Peer

Your PBIs are generally clear.

In PBI 1, “10–20 projects per page” is too broad for consistent testing, so defining a clear page size would improve precision and better align the PBIs with Scrum’s emphasis on testable requirements.

In PBI 3, “precisely locate” is subjective, so I suggest replacing it with measurable wording such as “match projects by selected criteria” and aligning AC1 with this general filtering rule rather than relying on a specific example.

Feedback Received

1. PBI 1 lacks the mandatory ”notes” functions specified in the requirements.
2. “Request list” AC in PBI 1 is suggested to shift to PBI 2 as the user story in PBI 1 only focuses on submitting the project, rather than checking history operations.
3. ACs are too detailed and atomic, try to combine them to focus on core business rules, rather than listing all the status changes.

5 Reflection on Learning

In this coursework, I gained a deeper understanding of the Scrum process. It organizes work around clear user goals and manageable scope, rather than simply combining related functions together. When defining Modules 5 and 6, our team’s discussion gave me an important insight. Initially, the definition of Module 6 included functions such as submission, status tracking, and request history. After discussion, we deliberately reduced the size of this module to ensure better coverage by PBIs. This made me realize that an effective PBIs design is not merely about combining similar functions, but rather maintaining clear responsibilities and ensuring that each module can be fully defined by PBIs.

The peer review further deepened my understanding. From the feedback I received, I realized that AC should cover a broader range of testable business rules rather than overly detailed steps. Moreover, PBIs need clearer boundaries to avoid overlap. Based on these feedbacks, I made adjustments, added missing remarks requirements, removed the request list function from PBID 1, and modified AC to make it less atomic and more in line with user goals. Overall, this process not only enhanced my ability in requirement analysis but also improved my ability to optimize work through collaboration.